

Fig. 4: Actual-size, single-side PCB for the remote

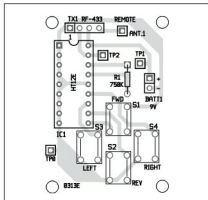


Fig. 5: Component layout for the PCB in Fig. 4

is avoided, the robot will continue moving forward.

To turn off the automatic mode, press S3 and S4 simultaneously. This algorithm can be further refined for different applications.

Software. The source code for the robot is written in 'C' language and compiled using MPLAB IDE. It is designed to run on Windows. Source program 'pic_rf.c' is used for the microcontroller. The program is well commented and easy to understand. The pic_rf.c file is converted into hex code using the MPLAB compiler. The hex code is burnt into the microcontroller using a suitable programmer.

Construction and testing

An actual-size, single-side PCB for the remote is shown in Fig. 4 and its component layout in Fig. 5. PCB for the robot is shown in Fig. 6 and its component layout in Fig. 7. Suitable

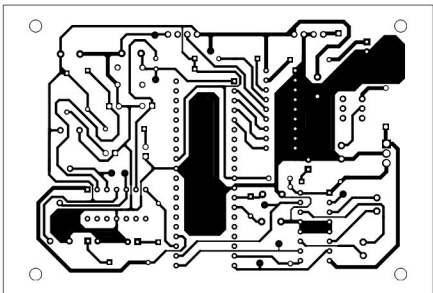


Fig. 6: Actual-size, single-side PCB for the robot

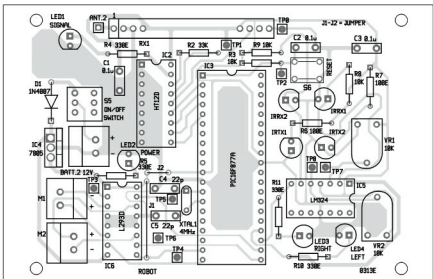


Fig. 7: Component layout for the PCB in Fig. 6

connectors are provided on both the PCBs to interface motors, RF modules and power supplies.

Testing of remote section. Check the correct input supply at TP1 with respect to TP0. Press any switch and monitor the changes at TP2 using an oscilloscope.

Testing of robot section. Verify

EFY Note

The source code of this project is included in this month's EFY DVD and is also available for free download on www.efymag.com website.

correct power supply to the robot at TP1 with respect to TP0. Proper reset to the microcontroller can be verified at TP2. Press any switch and check whether LED1 is glowing or not. Check the voltage at TP3, TP4, TP5 and TP6 to confirm whether correct actions are initiated by the microcontroller after receiving the code. LED3 and LED4 will glow if any obstacle comes in front of the robot. Reference voltage for the obstacle detection system can be checked and adjusted at TP7 and TP8. ●

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